



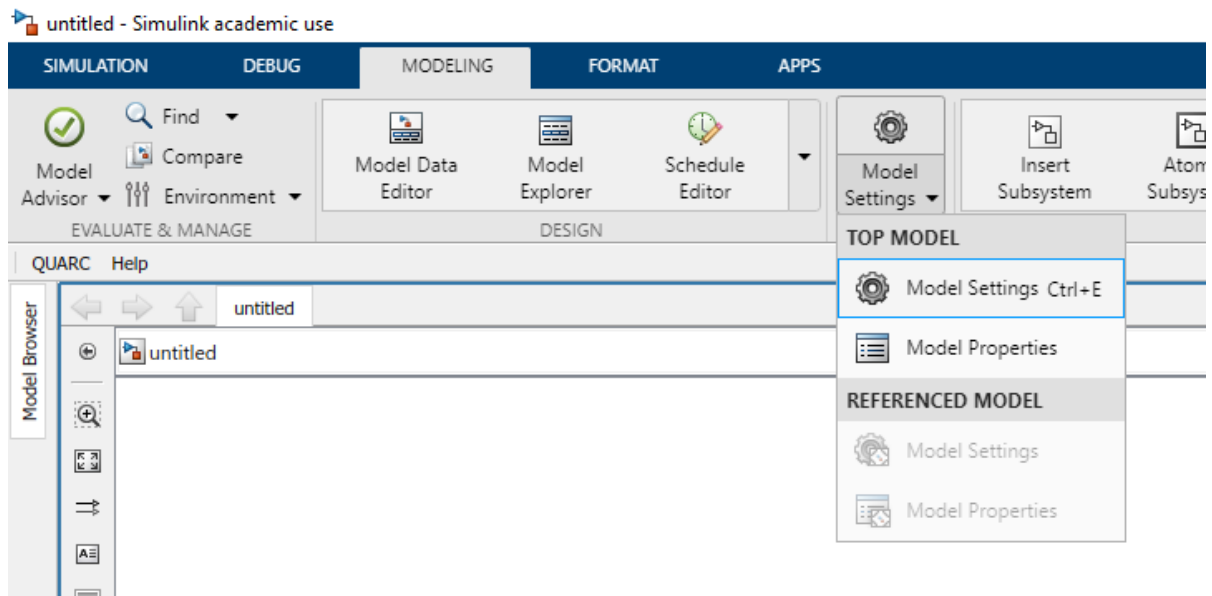
School of Engineering

Week 11 Laboratory Instructions

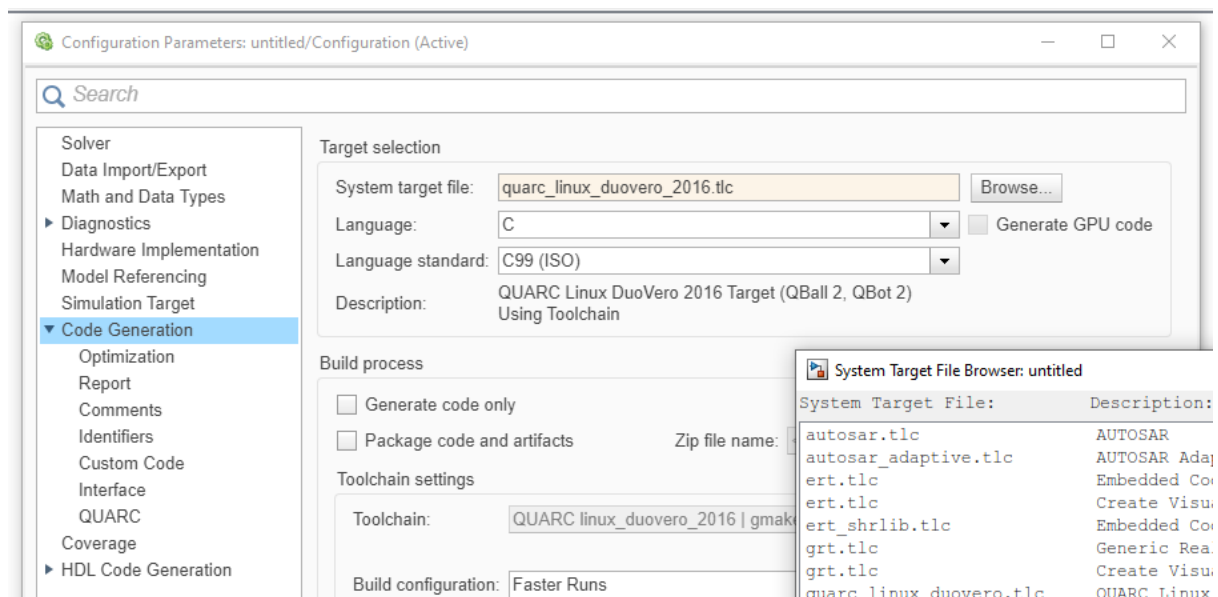
Autonomous Systems – MANU2206

Initial Preparations

1. Requirements:
 - a desktop computer, and
 - one QBot 2 which will be communicated and programmed through your desktop.
2. The username and password of the computer are *your RMIT student credentials*.
3. Make sure your PC and QBot 2 are powered ON.
4. Go to the *start* menu and search for *cmd* and click *Enter*. In the new black window (the Windows command window) that opens, type **ping 192.168.2.1**. This command is for pinging the router. If the connection is successful you will see the ping reply times being very short (usually less than 1 ms, but can be up to a few ms).
5. The QBot 2 can be pinged by typing *ping* (IP of the QBot 2) in the command window. Again, if the connection is successful, you will see the ping reply times being very short (usually not as short as what you see when pinging the router, may be in the order of tens to hundreds of milliseconds). If you cannot ping the robot, power cycle the QBot 2 (turn it off then on) and wait for about 60 seconds to reboot.
6. Open a blank model in Simulink or your built model (depending on the requirements of the task).
7. Go to the *MODELING* menu item and click on *Model Settings*.

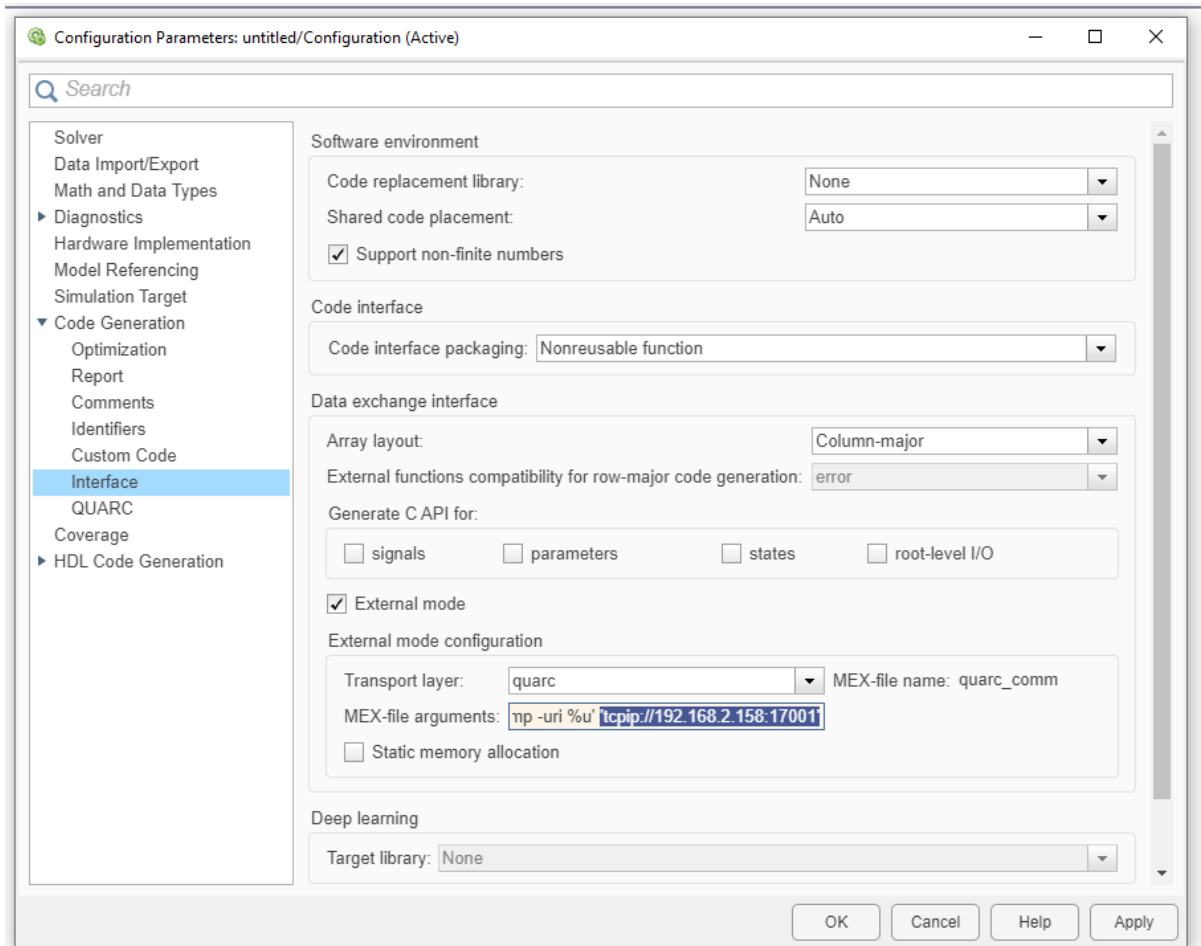


8. Under *Configuration Parameters*, under the tab *Code Generation*, find *System target file*. Click on *Browse*, choose *quarc_linux_duovero_2016.tlc*, click *Apply*, and then *OK*.



9. Under *Configuration Parameters/Code Generation*, under the tab *Interface*, find *MEX-file arguments*.

In the *MEX-file arguments* box, add the following at the end '*tcpip://192.168.2.158:17001*' where *192.168.2.158* is the IP address of a Qbot2. Change the last 3 digits of this IP address to the last 3 digits of your robot's IP address. You can find your robot's IP address printed on top of the PCB board. Click *Apply*, and *OK*.



Task 1: Turning various behaviours into modules

The aim of this task is to integrate multiple sensor y information in order to generate a behaviour.

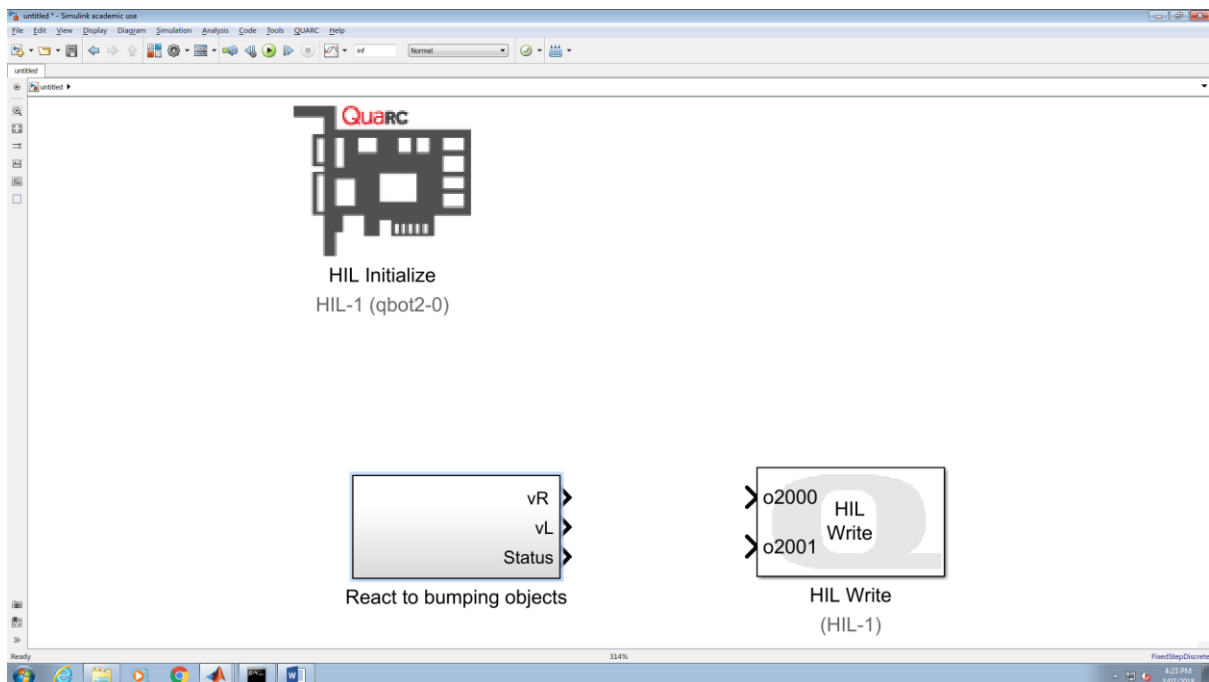
10. Open the **blank_model.xls** and save it as **W11_1.xls**.
11. Open the **W7_3.xls** model and copy its contents into **W11_1.xls**. Close the **W7_3.xls** model.
12. Double click on the MATLAB Function block and change its code as follows:

```
function [vR,vL,Status] = Controller(Bump_R,Bump_C,Bump_L)

vR = 0.2;
Status = 0;

if Bump_R || Bump_C || Bump_L
    vL = -0.2;
    Status = 1;
else
    vL = 0.2;
end
```

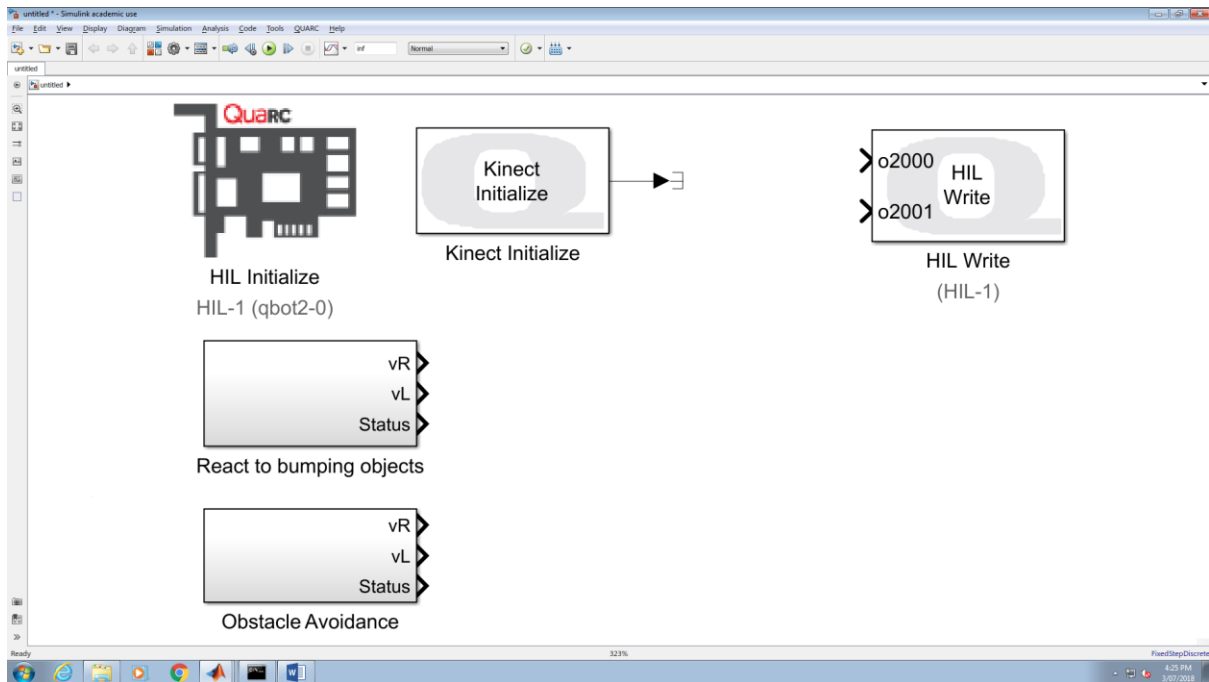
13. Select the *HIL Read* and the *MATLAB Function* and create a subsystem out of them. Name it “React to bumping objects”. Remove the connections to inputs of the *HIL Write* block.



14. Open the **W9_3.xls** model and copy its contents into **W11_1.xls**. Close the **W9_3.xls** model. In **W11_1.xls** remove the repeated blocks.
15. Double click on the *Controller* MATLAB Function you just copied, and replace its code with the following:

```
function [vR,vL,Status] = Controller(x,y,v,th)
Status = 0;
r = sqrt(x.^2+y.^2);
r = r(x>0);
if min(r)<th
    vR = v; vL = -v;
    Status = 1;
else
    vR = v; vL = v;
end
```

16. Create a subsystem of the *Kinect Get Depth*, the two MATLAB Functions, and the two Constant blocks. Name it “Obstacle Avoidance”.



17. Open the **W10_4.xls** model and copy its contents into **W11_1.xls**. Close the **W10_4.xls** model. In **W11_1.xls** remove the repeated blocks.
18. Double-click on the *Controller* MATLAB Function and replace its code with the following.

```
function [vr,vl,Status] = Controller(xc,A,V_A,V_R, Alimits)

Amin = Alimits(1);
Amax = Alimits(2);
Status = 0;

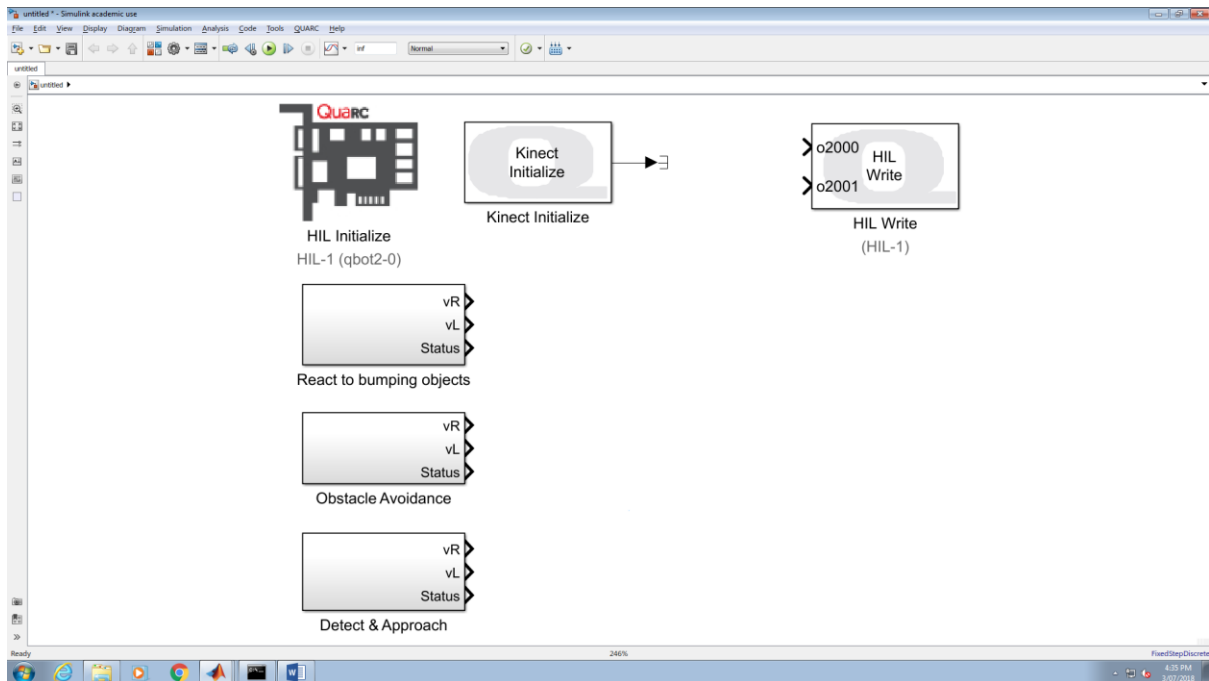
if A<Amin
    vr = 0.05;
    vl = -0.05;
elseif A>Amax
    vr = 0;
    vl = 0;
    Status = 1;
else
    Status = 1;
    if xc<20
        vr = V_R(2);
        vl = V_R(1);
    elseif xc>45
        vr = V_R(1);
        vl = V_R(2);
    else
        vr = V_A;
        vl = V_A;
    end
end
end
```

19. Select the following all together and create a subsystem:

- The *Detection* subsystem;
- The Constant blocks; and

- The Controller MATLAB Function.

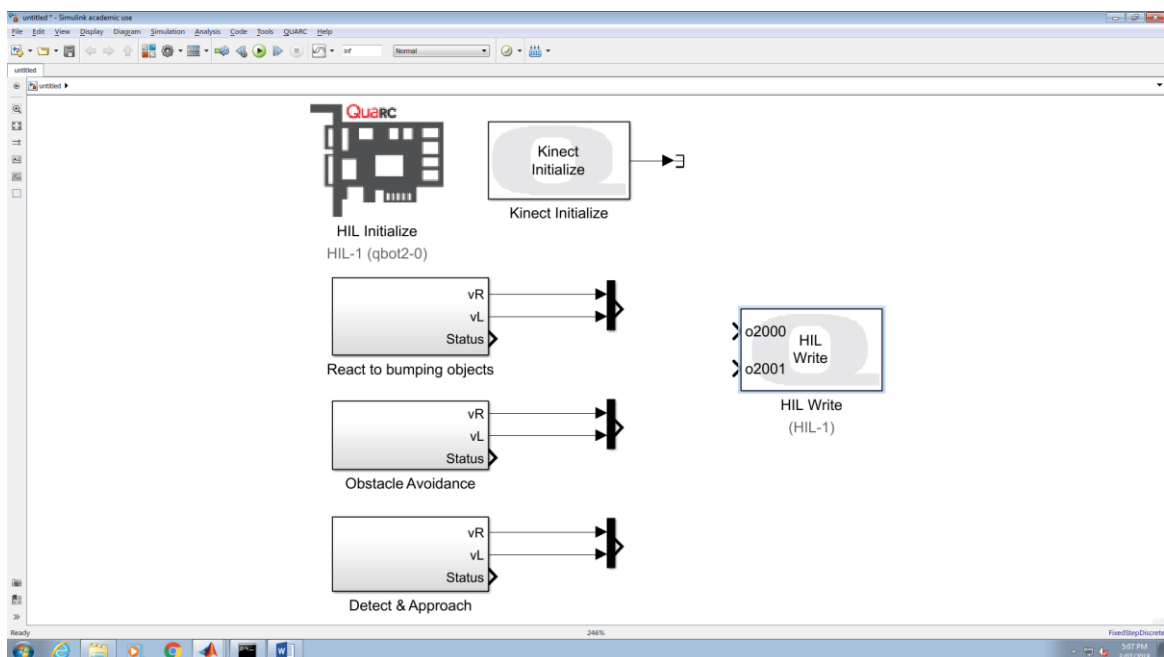
20. Name the new subsystem “Detect & Approach”.



21. Check the following blocks within the model and its subsystems, and replace their sample time with $qc_get_step_size * 10$

- Kinect Initialize
- HIL Write
- HIL Read (inside the “React to bumping objects” subsystem)
- Kinect Get Depth (inside the “Obstacle Avoidance” subsystem)
- Kinect Get Image (inside the “Detect & Approach” subsystem)

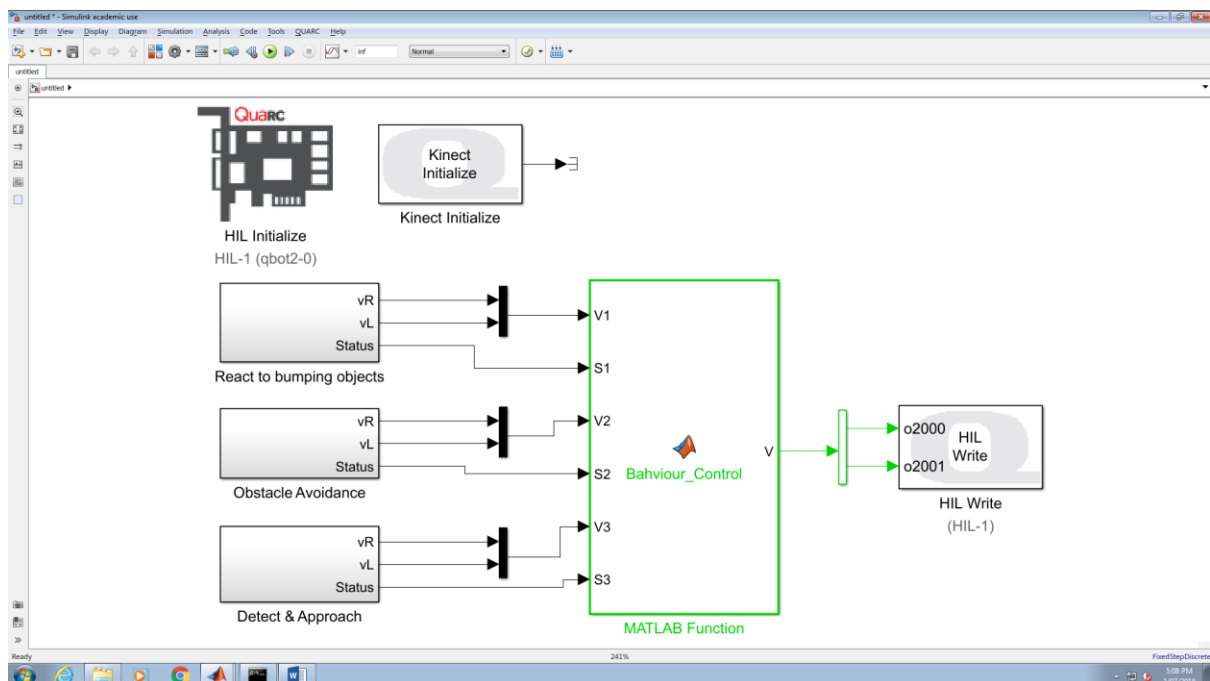
22. Add three **Mux** blocks and multiplex the vR and vL outputs of each of blocks.



23. Create a new MATLAB Function with the following codes in it:

```
function V = Bahviour_Control(V1,S1,V2,S2,V3,S3)
if S1
    V = V1;
elseif S2
    V = V2;
elseif S3
    V = V3;
else
    V = [0.2 0.2]';
end
```

24. Add a **demux** block to the model, then connect the blocks as shown below:



25. Save the model.

26. Compile and run the model. Understand how different behaviours are combined and selectively exhibited by the robot. How is the prioritisation set? What are the possible imperfections? Can you improve the *Behaviour_Control* function to make it work better?